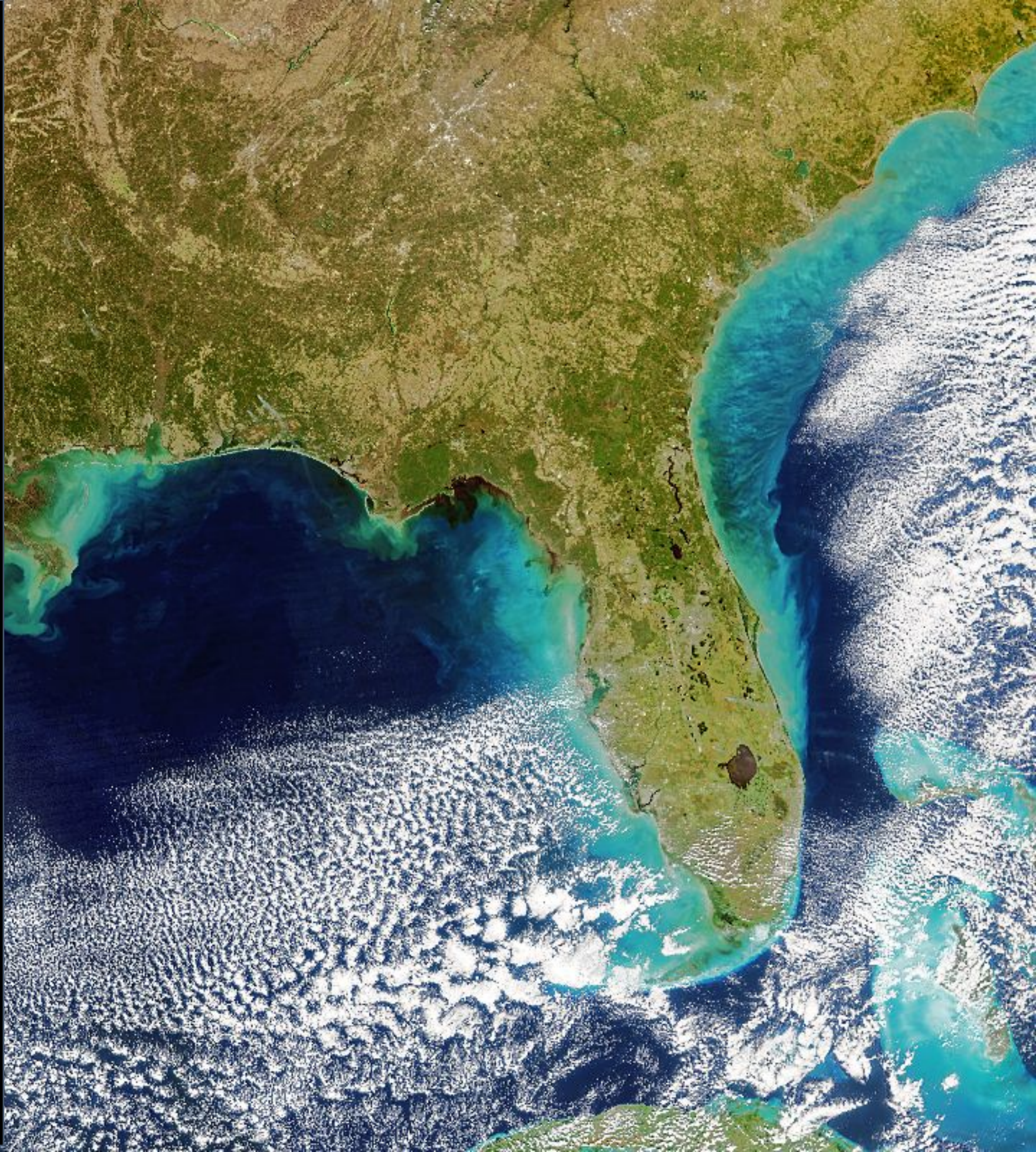
Introduction to PACE hyperspectral ocean color data for fisheries applications

Thanks to
Jeremy Werdell, PACE Project Scientist, NASA Goddard for PACE slides
and to NASA for images



Late 2014 PACE became a mission and work began early 2015

LAUNCH 8 February 2024

Pictures from NASA

NASA Plankton, Aerosol, Cloud, ocean Ecosystem (PACE) mission

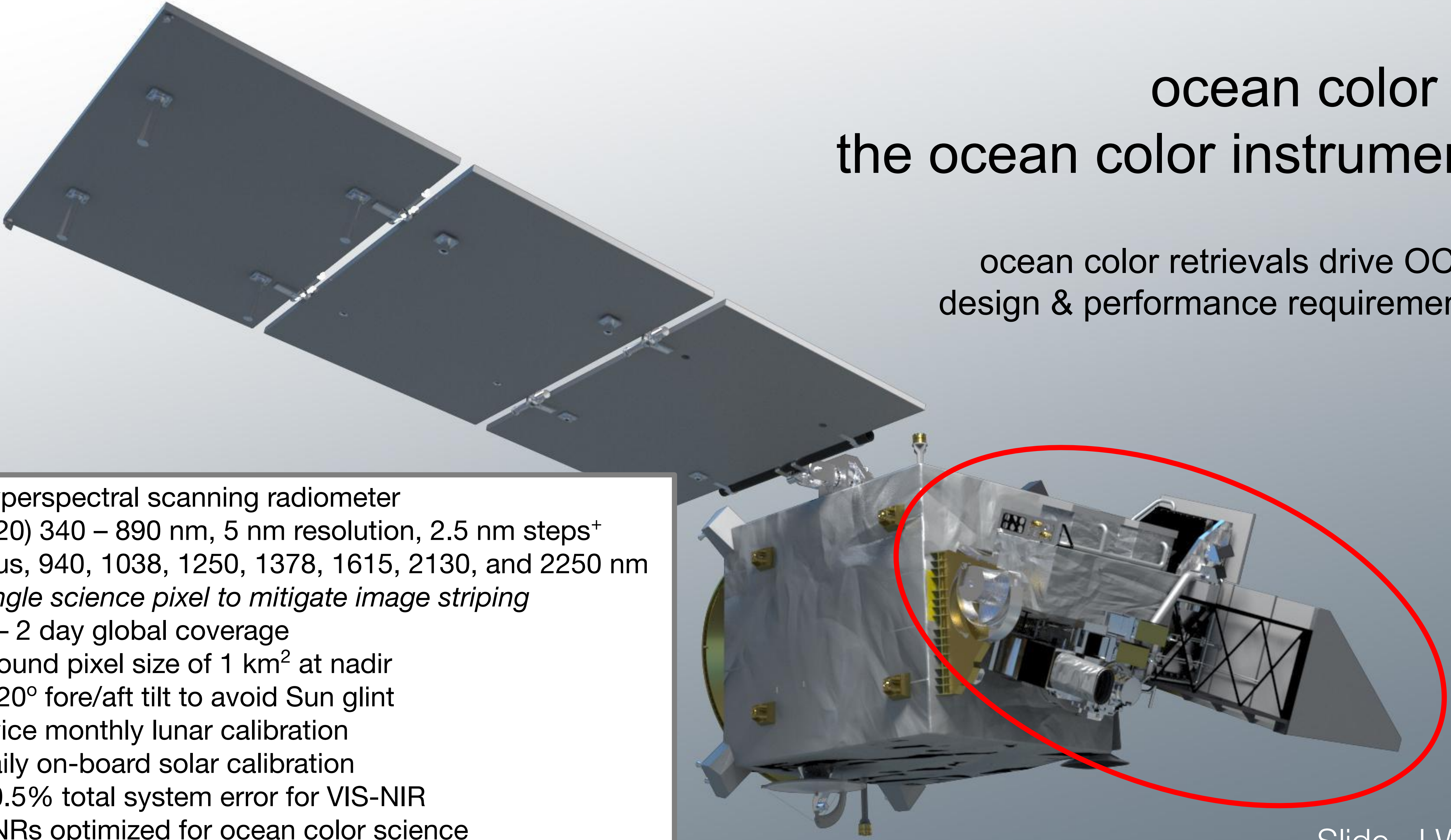
The **PACE (Plankton, Aerosol, Cloud, ocean Ecosystem)** satellite mission is a NASA Earth science project aimed at studying the Earth's ocean ecosystems and atmosphere. It will collect data on the health of the oceans, the composition of aerosols in the atmosphere, and the characteristics of clouds.

The sensors:

- Ocean Color Instrument (primary)
- SPEXone
- HARP2

ocean color & the ocean color instrument

ocean color retrievals drive OCI's design & performance requirements

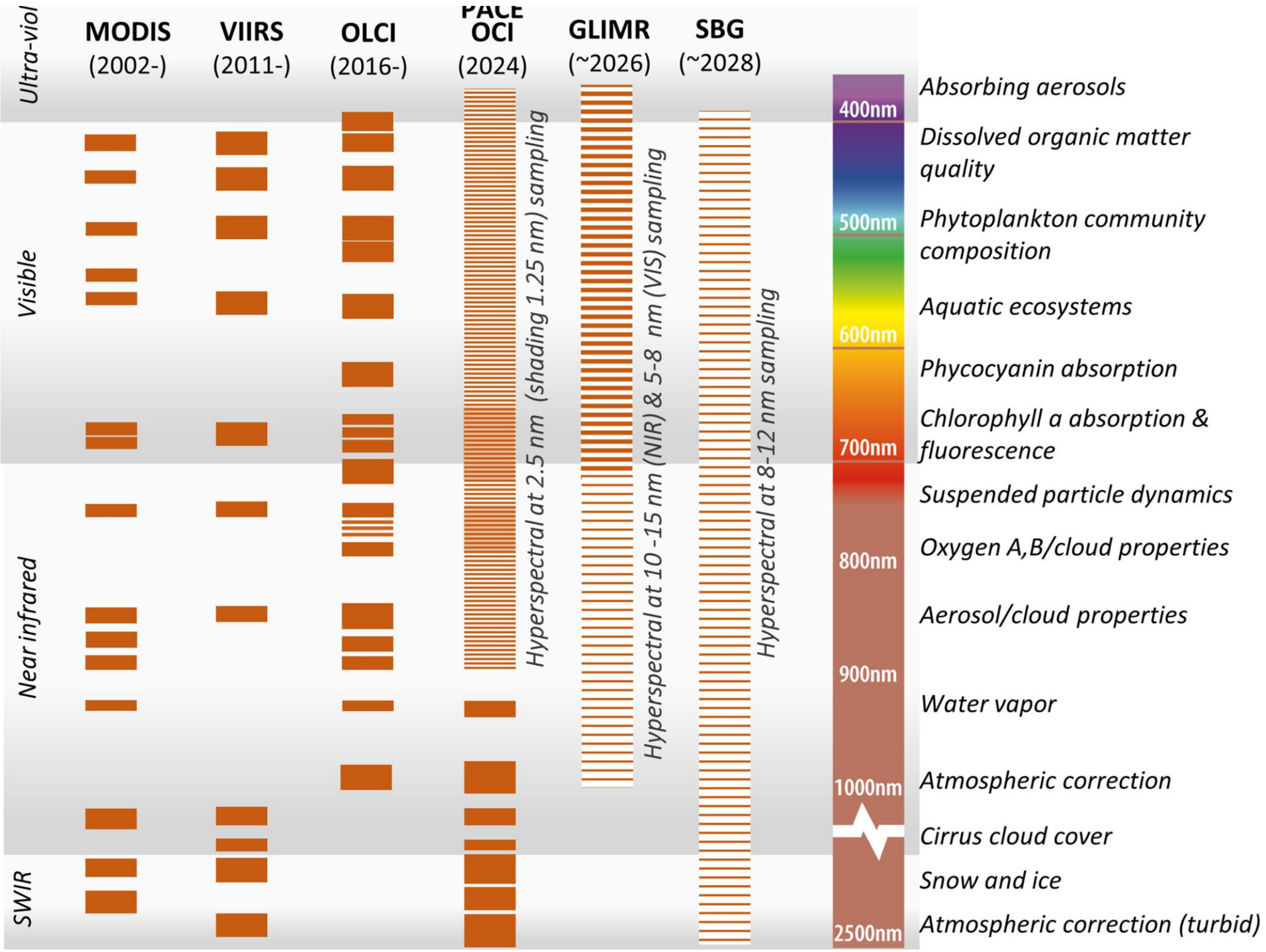


- hyperspectral scanning radiometer
- (320) 340 – 890 nm, 5 nm resolution, 2.5 nm steps^{+}
- plus, 940, 1038, 1250, 1378, 1615, 2130, and 2250 nm
- *single science pixel to mitigate image striping*
- 1 – 2 day global coverage
- ground pixel size of 1 km^2 at nadir
- ± 20° fore/aft tilt to avoid Sun glint
- twice monthly lunar calibration
- daily on-board solar calibration
- <0.5% total system error for VIS-NIR
- SNRs optimized for ocean color science
- simulated top-of-atmosphere data available

+ with 1.25 nm steps in several spectral regions
* developed primarily for mechanical processing assessments

Slide, J Werdell

Hyperspectral allows us to distinguish what is in the water

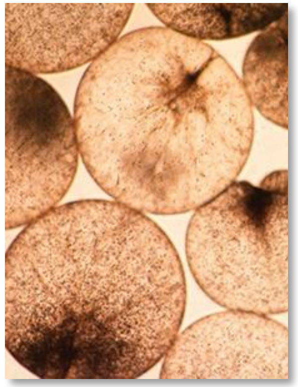


Comparison of the spectral sampling of different missions. Figure from "Dierssen, H. M., Gierach, M., Guild, L. S., Mannino, A., Salisbury, J., Schollaert Uz, S., et al. (2023). Synergies between NASA's hyperspectral aquatic missions PACE, GLIMR, and SBG: Opportunities for new science and applications. *Journal of Geophysical Research: Biogeosciences*, 128, e2023JG007574. https://doi.org/10.1029/2023JG007574"

Example diatom

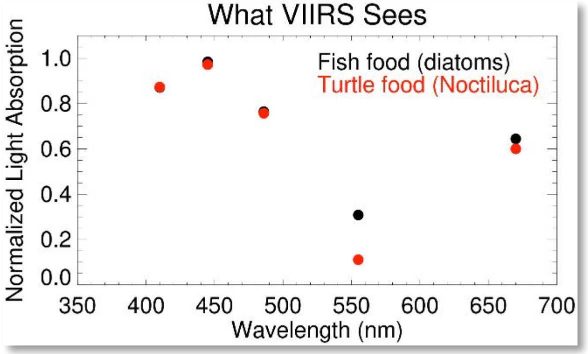
Linda Armbrecht, abc.com.au

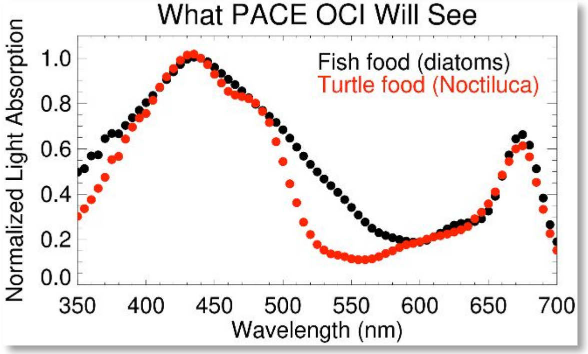
Example Noctiluca



1 mm

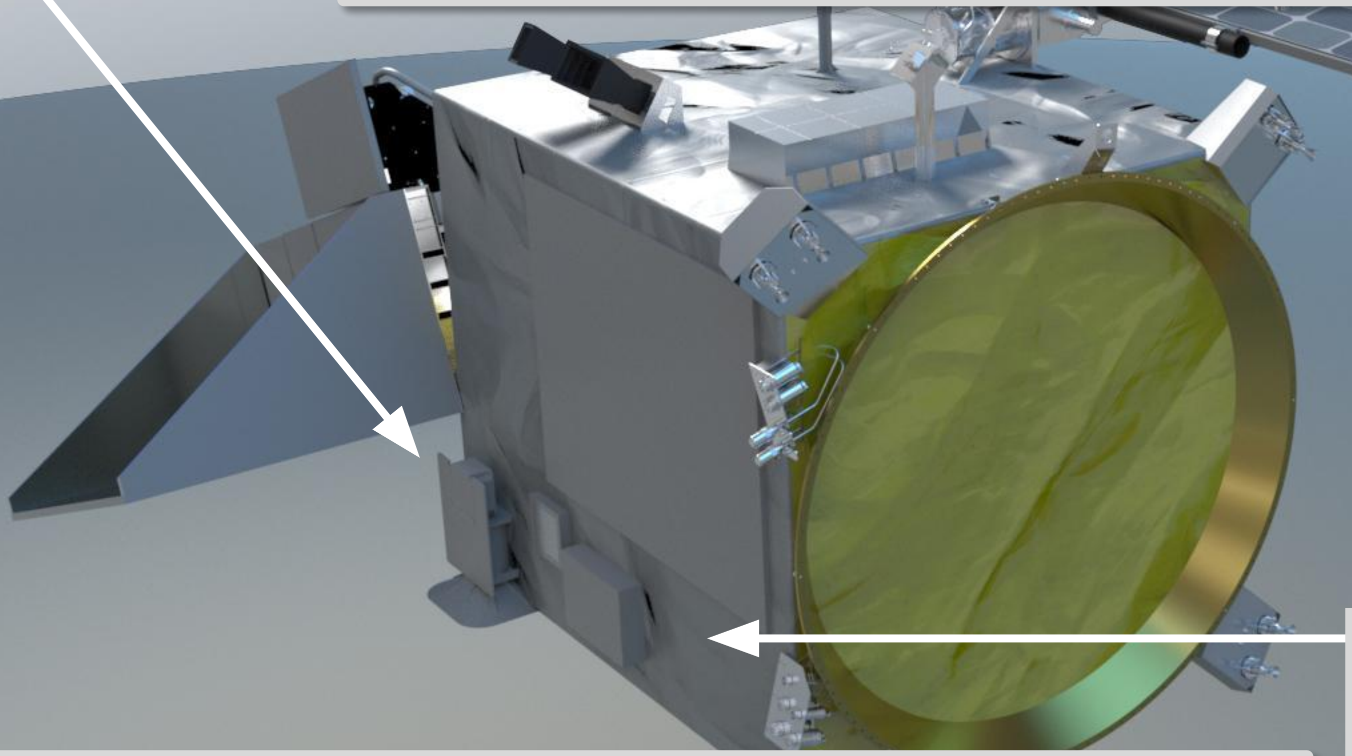
Joaquim Goes, LDEO





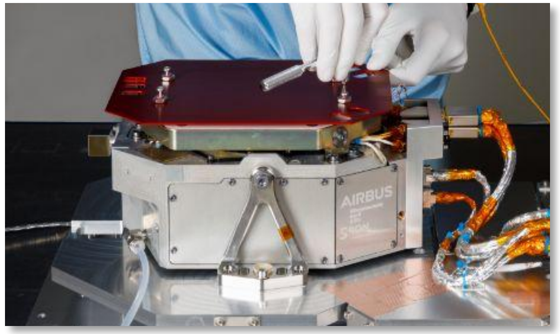
UMBC Hyper Angular Rainbow Polarimeter (HARP-2)

- Excellent for cloud droplet size and ice particle shape/roughness retrievals
- *Wide swath matches OCI, offering potentially improved atmos. Correction*
- *Physical properties of aerosols: size, shape, distribution vertically and horizontally, how much light do they block (opaqueness)*



Slide modified from,J Werdell

SRON/Airbus Spectro-polarimeter for Planetary Exploration (SPEXone)



- Aerosol characterization
- *What are they made of? Sulfates, dusk, smoke, pollution etc*
- *How do they affect light? Refraction, absorbtion, scattering*

https://pace.oceansciences.org/data_table.htm

Data Products Table

Calibrated Radiometry and Polarimetry | Aquatic Properties to be Produced by OCI | Atmospheric Properties to be Produced by OCI | Land Data Products to be Produced by OCI | Aerosol and Ocean Properties from HARP2 | Aerosol and Land Surface Properties from HARP2 | Cloud Properties from HARP2 | Aerosol, Ocean, and Land Properties from SPEXone | Aerosol and Ocean Properties from OCI + HARP2 + SPEXone

Points of contact (POCs) under Additional Info refer to the PACE Project Science Lead responsible for shepherding an approach through implementation.

Access to data varies with its status (data maturity level). Provisional data are available through Earthdata Search, the OB.DAAC File Search and Level 3 & 4 Browser. Test and Diagnostic data are available through the OB.DAAC File Search and Level 3 & 4 Browser. See also "Access PACE Data".

What do colors in the "Availability" column mean?

Available	Coming soon!	Currently implementing and evaluating	No approach currently identified

Calibrated Radiometry and Polarimetry

Calibrated and geolocated radiometry and polarimetry as observed at sensor.

Product	L2 Suite	Description and Use	Units	Availability	Status	Additional Info
Spectral top-of-atmosphere radiances from OCI	*N/A*	Spectral radiance observed at the top of the atmosphere.	W m^{-2} um^{-1} sr^{-1}	Level-1B 1.2-km native; daily - Level-1C; 5.2-km; daily	Provisional	Level-1C draft data format and examples
Spectral top-of-atmosphere radiances and polarimetry from SPEXone	*N/A*	Spectral radiance and polarimetry observed at the top of the atmosphere, for all sensor viewing angles.	Various	Level-1B 5.2-km; daily - Level-1C; 5.2-km; daily	Provisional	Level-1C draft data format and examples
Spectral top-of-atmosphere radiances and polarimetry from	*N/A*	Spectral radiance and polarimetry observed at the top of the atmosphere.	Various	Level-1B 5.2-km; daily - Level-1C; 5.2-km; daily	Provisional	Level-1C draft data format and examples

Levels

Level 1: Calibrated raw data from sensors. Top of atmosphere radiances from OCI, HARP2, and SPEXone. OCI measures Rrs (Remote Sensing Reflectance): color intensity of the ocean when viewed from above.

Level 2: Derived geophysical variables at the original sensor resolution. Products like remote sensing reflectance (Rrs), absorption properties (derived from Rrs), aerosol properties, cloud properties, chlorophyll estimates (derived from absorption), phytoplankton composition

Level 3: Gridded global maps of those variables, averaged over time (daily, monthly, etc.)

Products

Absorption coefficients: how light is absorbed *within* the water by different substances (like phytoplankton or dissolved organic matter). Derived from Rrs

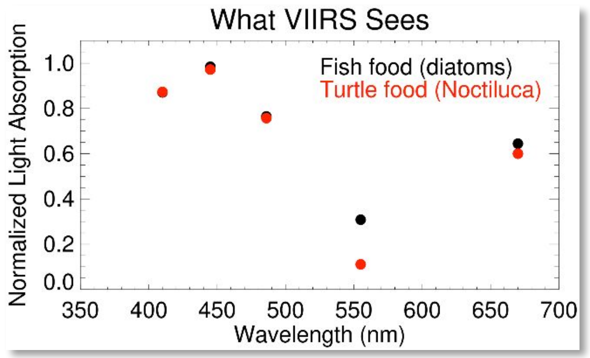
Chlorophyll-a: estimate from empirical equations involving Rrs or absorption coefficients

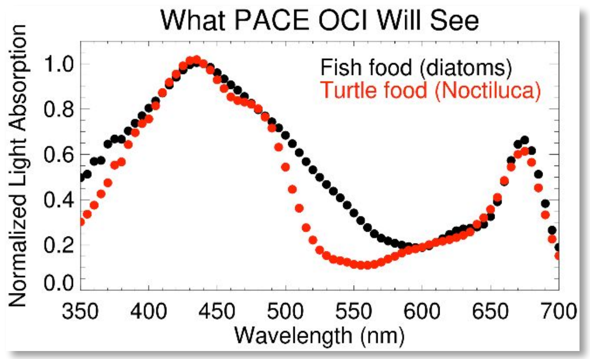
Phytoplankton species and community composition: new to PACE via the hyperspectral absorption coefficients and backscattering coefficients

Diffuse Absorption coefficients (Kd): how quickly light decreases with depth in water at a given wavelength

Spectral phytoplankton absorption coefficients: how much light is absorbed by phytoplankton pigments in water

Phytoplankton carbon: a more direct indicator of phytoplankton biomass





PACE Products for Fisheries, Ryan Vandermeulen (NOAA Fisheries)

Applications of PACE hyperspectral ocean color data for aquaculture and fisheries management

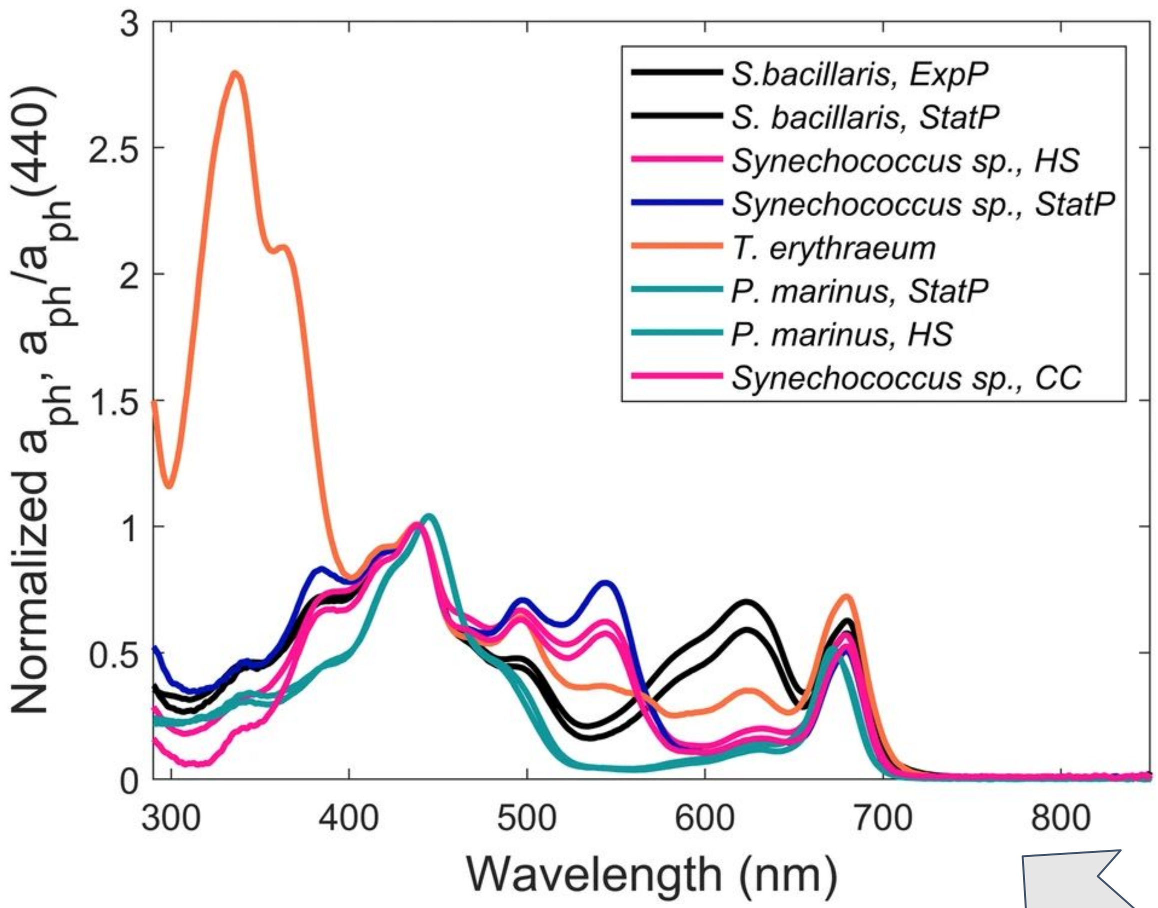
AUTHOR
Ryan Vandermeulen, Veronica Lance, Clete Otoshi, Ken Riley, Michelle Tomlinson, and Rebecca Trinh; Python/R code by Elizabeth Holmes

PUBLISHED
2025-04-01T00:00

This is a condensed version of NOAA Tech. Memo. NMFS-F/SPO-253.

PACE Ocean products tell you about:

- WHAT IS THE COLOR OF THE OCEAN (Rrs)
- WHAT/WHO IS THERE (phytoplankton composition, SPM)
- HOW MUCH OF IT IS THERE (chlorophyll concentration, particulate/phytoplankton carbon etc.)
- ARE THEY SHINY/COLORFUL??? (IOPs)
- HOW MUCH LIGHT IS THERE (PAR/Kd)
- WHAT ARE THEY DOING (Primary productivity)
- HOW ARE THEY DOING (Fluorescence Line Height)



ELSEVIER

Remote Sensing of Environment

Volume 302, 1 March 2024, 113964

Review

Phytoplankton composition from sPACE: Requirements, opportunities, and challenges

Ivona Cetinić [a b 1], Cecile S. Rousseaux [b 1], Ian T. Carroll [c b], Alison P. Chase [d], Sasha J. Kramer [e], P. Jeremy Werdell [b], David A. Siegel [f], Heidi M. Dierssen [g], Dylan Catlett [h], Aimee Neeley [b i], Inia M. Soto Ramos [a b], Jennifer L. Wolny [j], Natasha Sadoff [b i], Erin Urquhart [b i], Toby K. Westberry [k], Dariusz Stramski [l], Nima Pahlevan [m i], Bridget N. Seegers [a b], Emerson Sirk [b i], Priscila Kienteca Lange [n o] ...Michael Sayers [t]

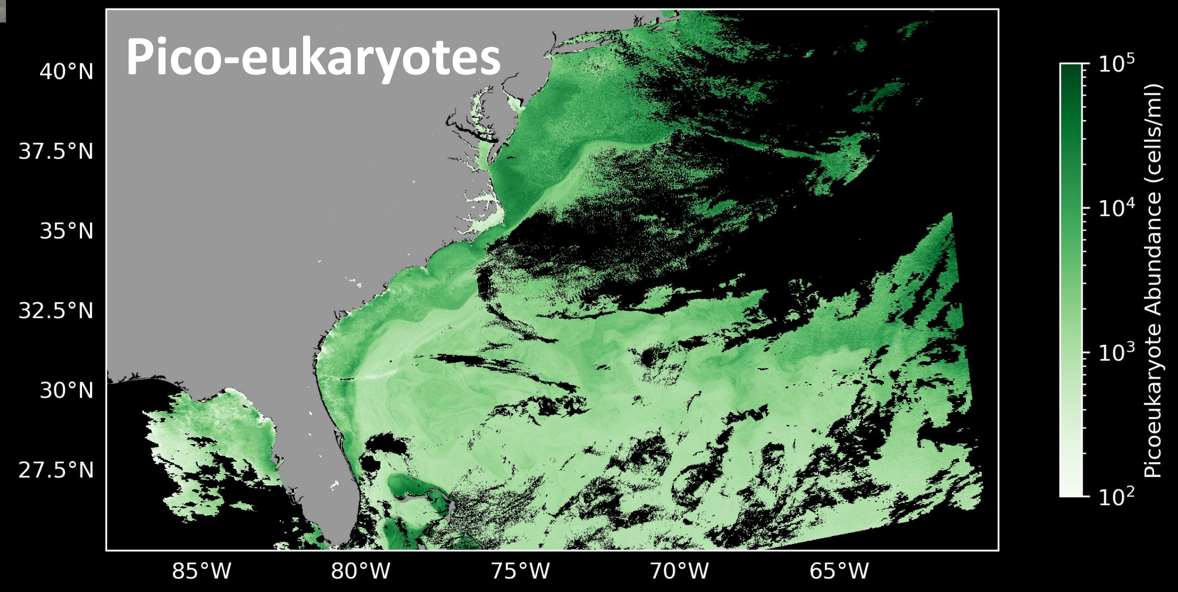
Data Descriptor | Open access | Published: 03 February 2024

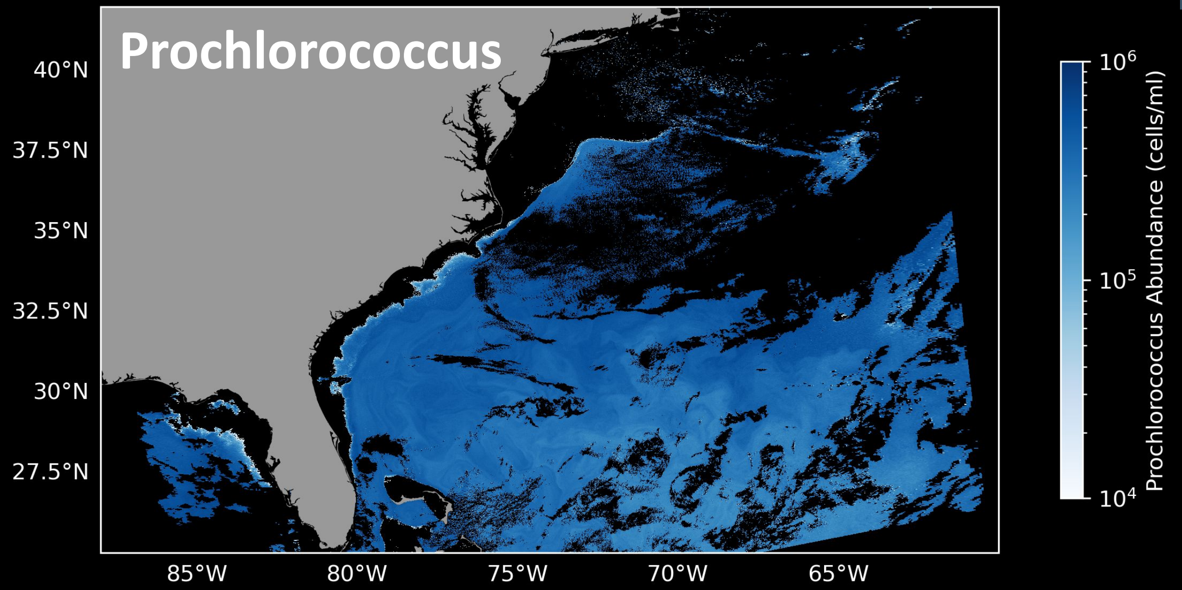
Phytoplankton optical fingerprint libraries for development of phytoplankton ocean color satellite products

Michael W. Lomas, Aimee R. Neeley, Ryan Vandermeulen, Antonio Mannino, Crystal Thomas, Michael G. Novak & Scott A. Freeman

Scientific Data **11**, Article number: 168 (2024) | Cite this article

PACE OCI Phytoplankton Community Composition



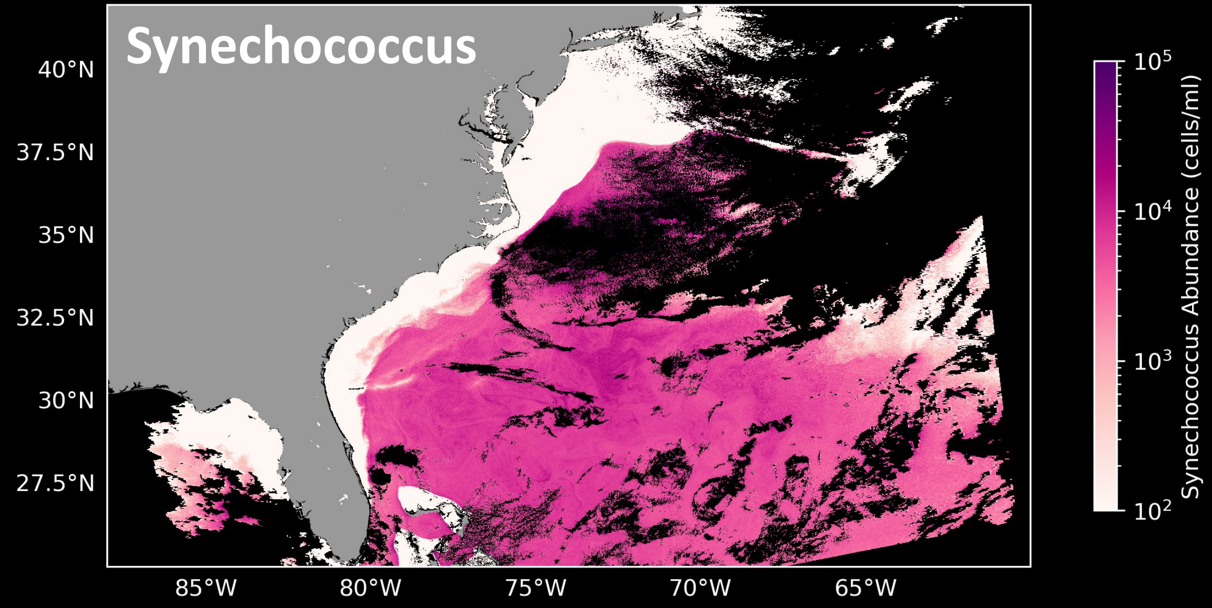


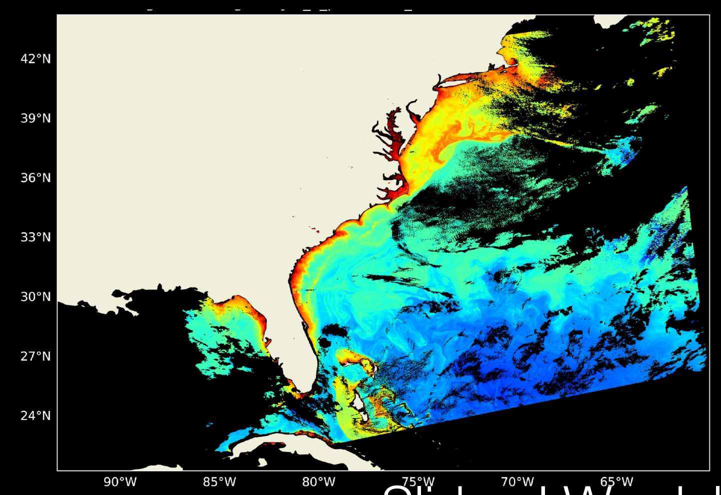
21 March 2024

chlorophyll-a

phytoplankton community

MOANA algorithm
Lange et al. 2020

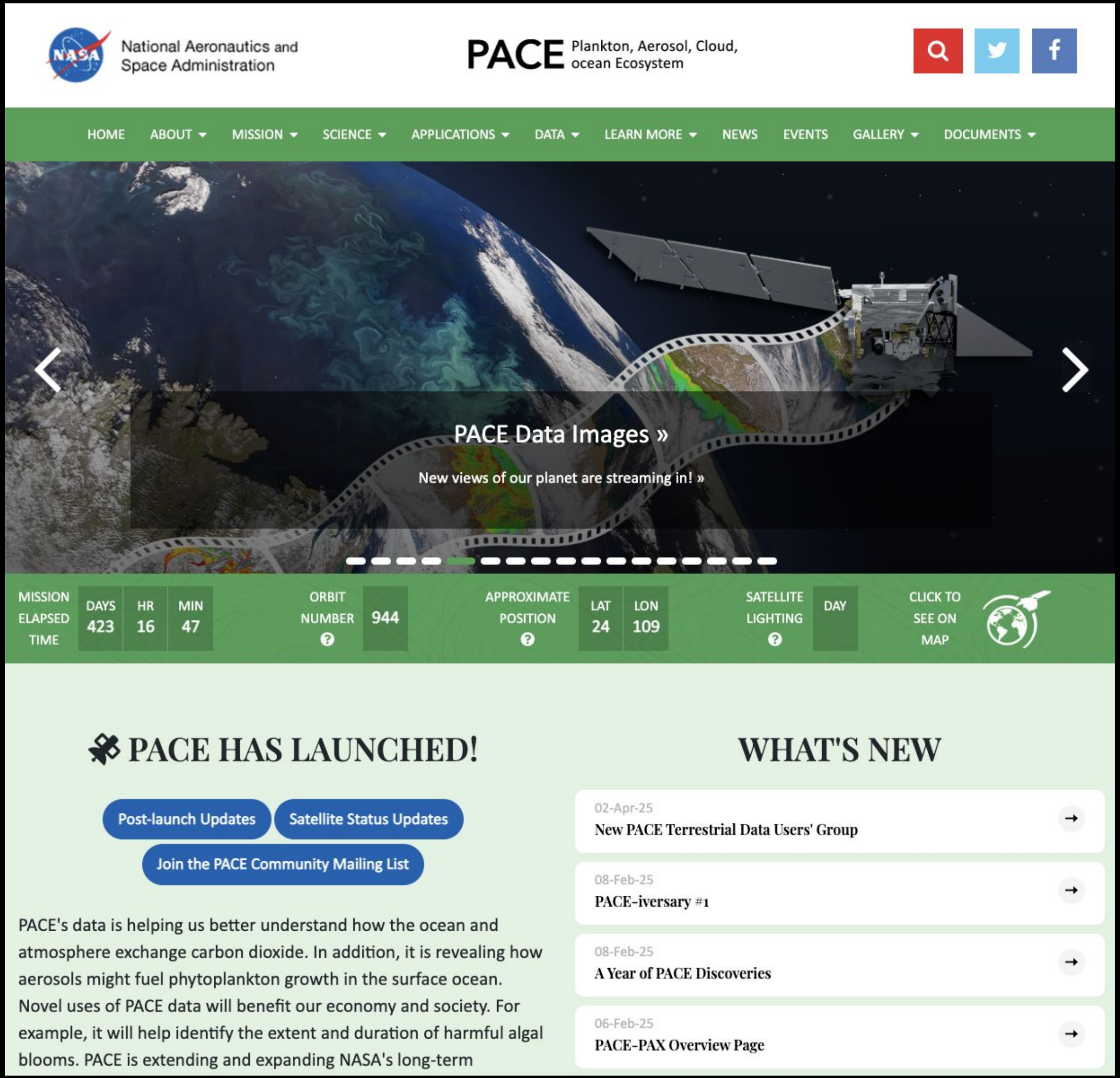


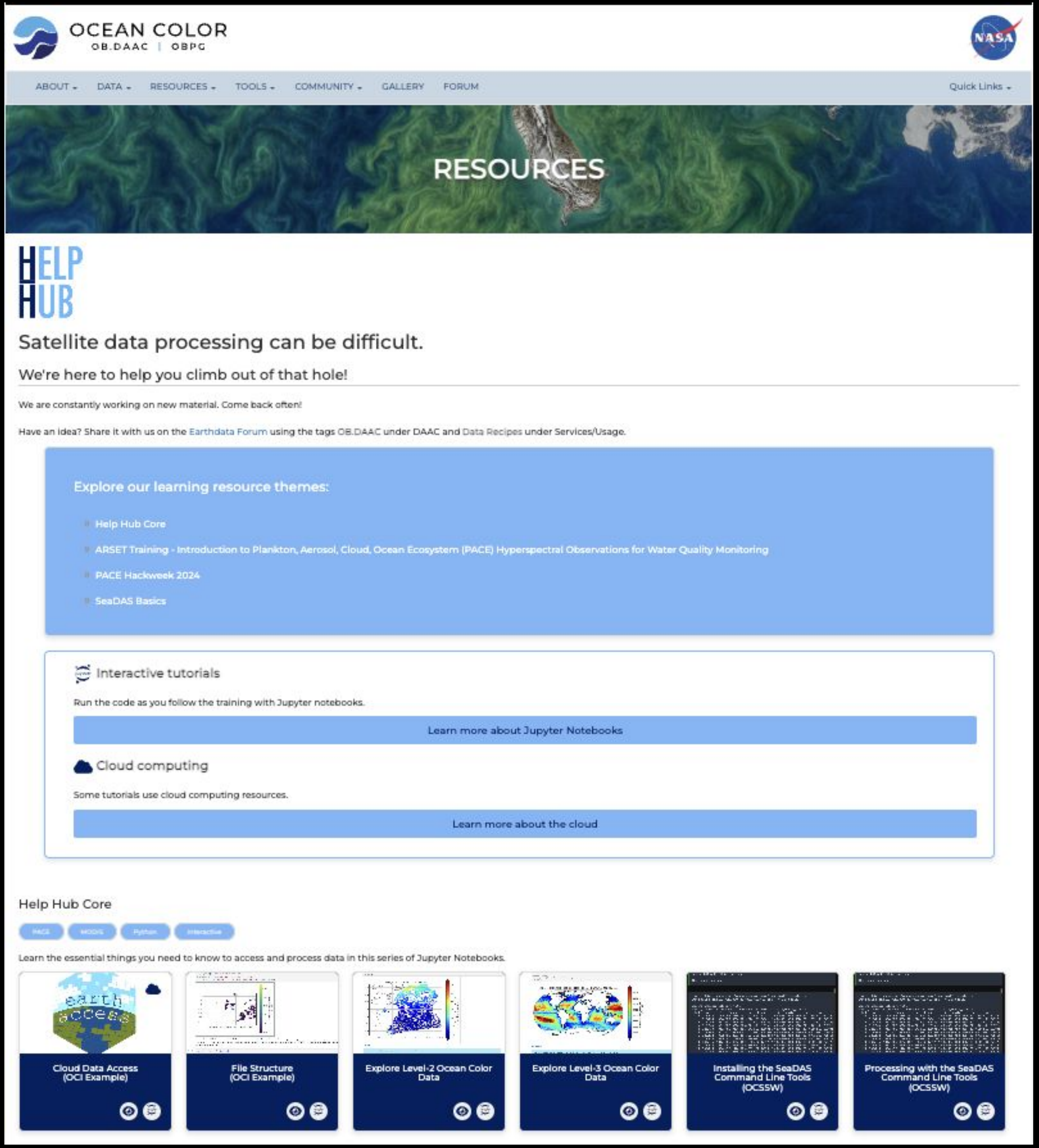


Slide, J Werdell

https://pace.gsfc.nasa.gov

https://oceancolor.gsfc.nasa.gov/resources/docs/tutorials/





Accessing PACE Data

Earthdata Search OB.DAAC portal

Earthdata Search OB.DAAC portal

https://search.earthdata.nasa.gov/search

Top left click “Browse Portals”
Click on “OBDAAC”
Filter Instruments to “OCI”

OB.DAAC Search page

https://oceandata.sci.gsfc.nasa.gov/

How can I access PACE data that’s in the cloud?

You can directly download netcdfs and run analysis. But these files are really big…often better to just get the subset of data that you want.

- OPeNDAP servers allow this so you download or access what you need
- or we can work with data in cloud buckets using earthaccess Python package

Resources

- Applications of PACE hyperspectral ocean color data for aquaculture and fisheries management, R Vandermeulen et al.

- PACE HackWeek 2024 https://pacehackweek.github.io/pace-2024/

- ARSET Course https://appliedsciences.nasa.gov/sites/default/files/2024-10/PACE_Part2_Final_0.pdf

- NOAA HackDays https://nmfs-opensci.github.io/NMFSHackDays-2025/

- EDMW PACE Workshop R Vandermeulen and E Holmes